

(TR-104) PYTHON WITH AI/ML

Training Day 52 Report

Date: 24 March 2026

Topic Covered: Revision of Previous AI/ML and NLP Concepts

The fifty-second day of training focused on revising previously learned concepts related to Transformers, RAG pipelines, fine-tuning, Streamlit, and AI application development to strengthen overall understanding and prepare for upcoming advanced topics.

- Revised Transformer architecture and concepts including self-attention, contextual embeddings, encoder-decoder models, and hidden states.
- Revised fine-tuning workflows for Transformer-based models using Hugging Face.
- Reviewed PEFT and LoRA concepts along with their advantages over full fine-tuning.
- Revised RAG (Retrieval-Augmented Generation) pipeline architecture and document retrieval workflow.
- Reviewed vector embeddings, vector databases, chunking strategies, and semantic search concepts.
- Revised guardrails and validation techniques for secure AI applications.
- Reviewed FastAPI concepts including API routes, request handling, middleware, and deployment workflow.
- Revised Streamlit concepts including widgets, session state, caching, forms, layouts, and deployment workflow.
- Reviewed integration workflow between Streamlit frontend, FastAPI backend, and Ollama models.
- Strengthened understanding of production-ready AI application architecture and optimization techniques.
- Revised concepts related to model serving, caching, logging, and response optimization in AI systems.